

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

Roll No. : S033	Name: Rudra Rawal
Class : SYIT	Batch : 1
Date of Assignment : 2026-07-15	Date/Time of Submission : 2026-07-15

Queue Application:

Write a program to:

- a. Implement a queue using an array.

CODE:

```
Practical_4A.py - C:/Users/mvlui/Desktop/SYIT_033/DS/Practical_4A.py (3.14.6)
File Edit Format Run Options Window Help

import array as arr
que = arr.array('i', [])

si = 3
print(que)

def enqueue(data):
    if(len(que) == si):
        print("Queue Overflow")
    else:
        que.append(data)
        print(que)

def dequeue():
    if(len(que) == 0):
        print("Queue Underflow")
    else:
        que.pop(0)
        print("Data removed: \n", que)

def peek():
    if(len(que) == 0):
        print("Queue is Empty")
    else:
        print("First element is: ", que[0])

def show():
    if(len(que) == 0):
        print("Queue is Empty")
    else:
        print("---- QUEUE ---- \n", que)

dequeue()
peek()
enqueue(10)
enqueue(11)
enqueue(12)
peek()
show()
enqueue(13)

print("Rudra_S033")
```

OUTPUT:

```
array('i')
Queue Underflow
Queue is Empty
array('i', [10])
array('i', [10, 11])
array('i', [10, 11, 12])
First element is: 10
---- QUEUE ----
array('i', [10, 11, 12])
Queue Overflow
Rudra S033
```

- b. Simulate a simple queuing system (e.g., customer service queue).

CODE:

```
*Practical_4B.py - C:\Users\mvlui\Desktop\SYIT_033\DS\Practical_4B.py (3.14.6)*
File Edit Format Run Options Window Help
import time

que = [("C1",2), ("C2",5), ("C3",7), ("C4",0)]

print(" ----- Customer Order ----- ")
print(que)

while que:
    Order, timer = que.pop(0)
    print(f" Serving: {Order} {timer} sec")
    time.sleep(timer)

print(que)

print("Rudra_S033")
```

OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2024; v.1944 64 bit (AMD64)) on win32
Enter "help" below or click "Help" above for
>>>
===== RESTART: C:\Users\mvlui\Desktop\SYIT_033\DS\Practical_4B.py =====
----- Customer Order -----
[('C1', 2), ('C2', 5), ('C3', 7), ('C4', 0)]
Serving: C1 2 sec
Serving: C2 5 sec
Serving: C3 7 sec
Serving: C4 0 sec
[]
Rudra_S033
```

Curiosity Questions:

1. Where do you see queues in daily life?
2. Why is **FIFO (First In First Out)** fair in real life? Can you think of situations where it's *not* fair?
3. What will happen if you try to **enqueue more elements than the queue size**?
4. What if you **dequeue from an empty queue**?
5. Imagine you're designing a **hospital emergency system**. Should it use a simple queue or something else? (hint: priority queue).
6. If **5 customers arrive at the same time**, but the service desk can serve only 2 at once, how would you design that?